



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

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- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

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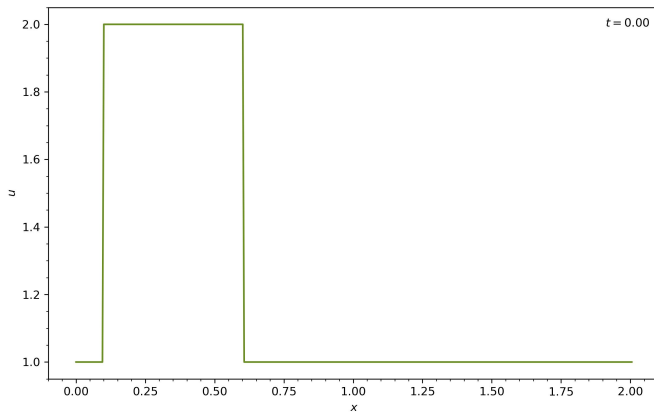
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Linear Advection

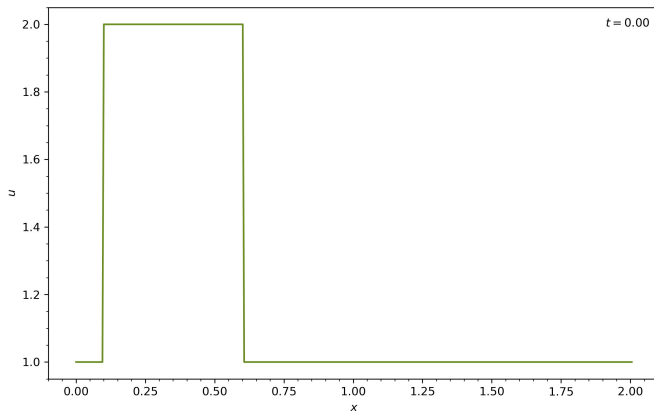
Linear Advection

■ *Initialization:*



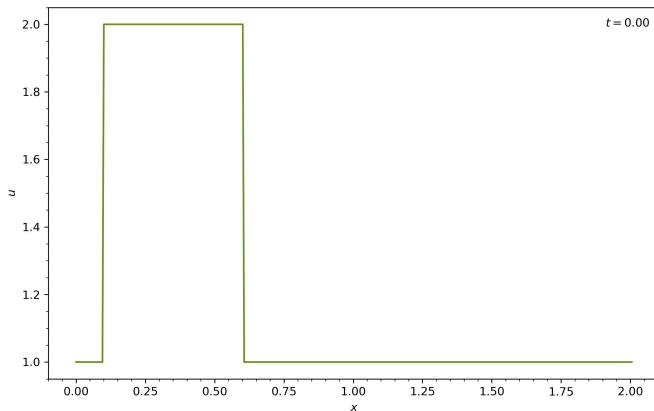
Linear Advection

- *Initialization:*
 - square wave



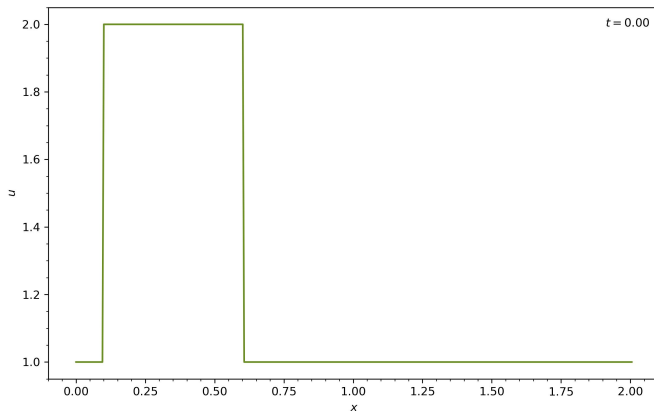
Linear Advection

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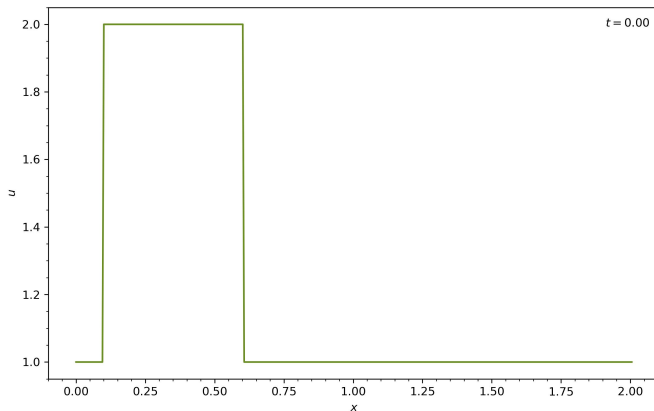
Linear Advection

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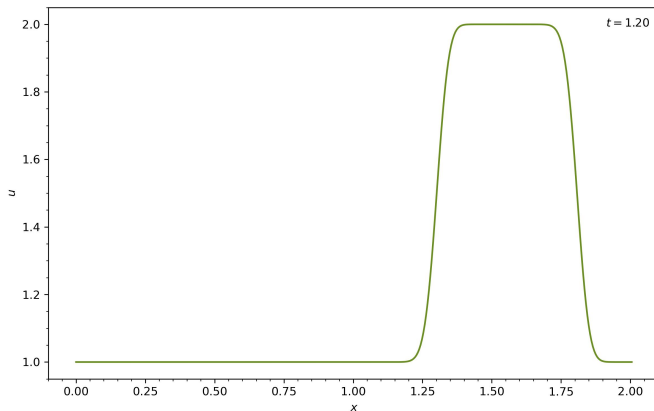
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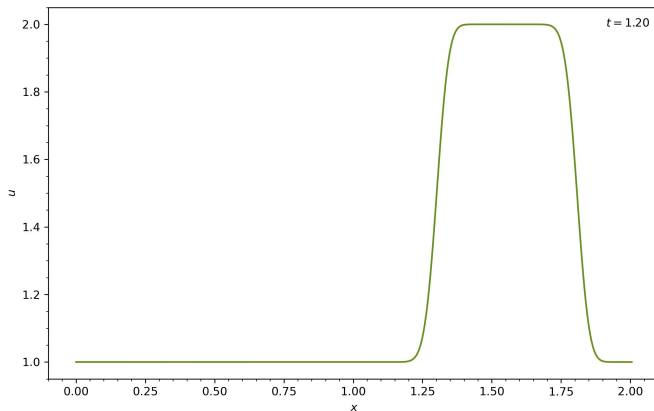
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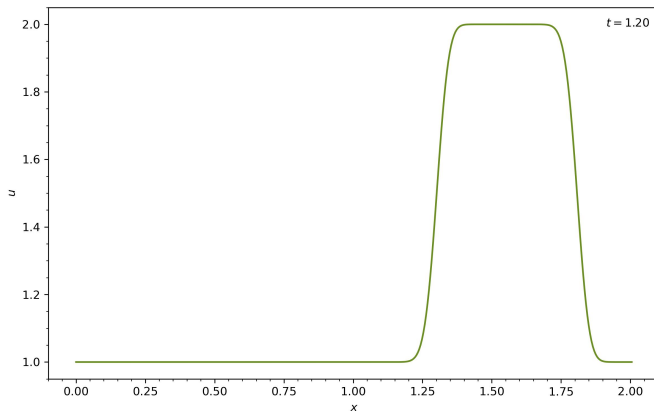
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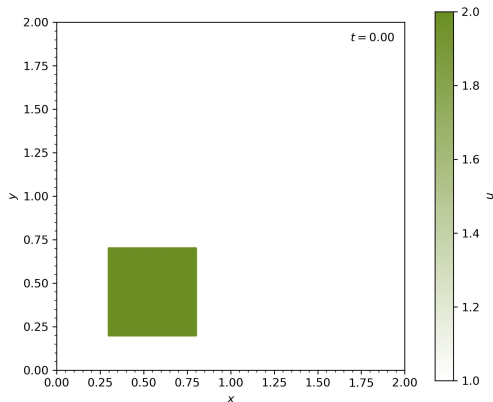
Linear Advection

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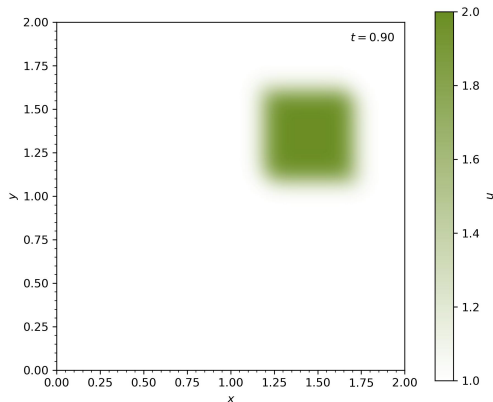
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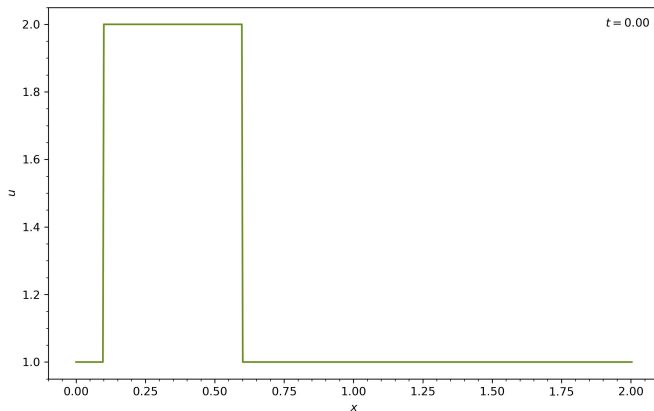
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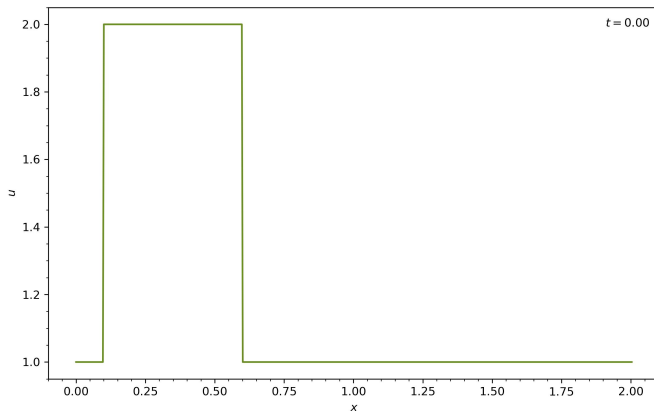
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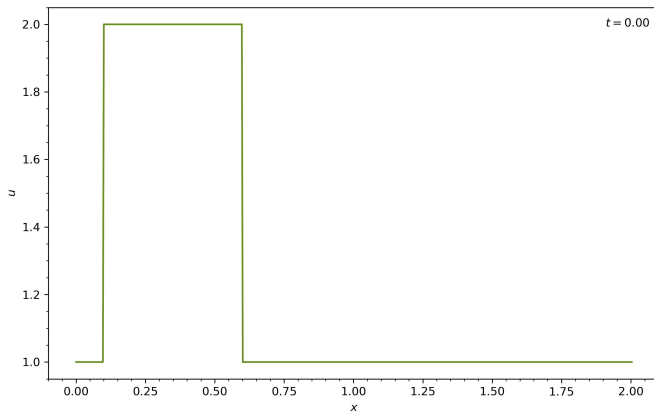
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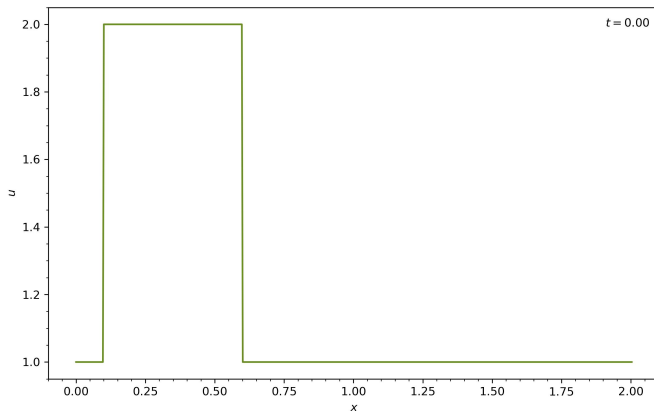
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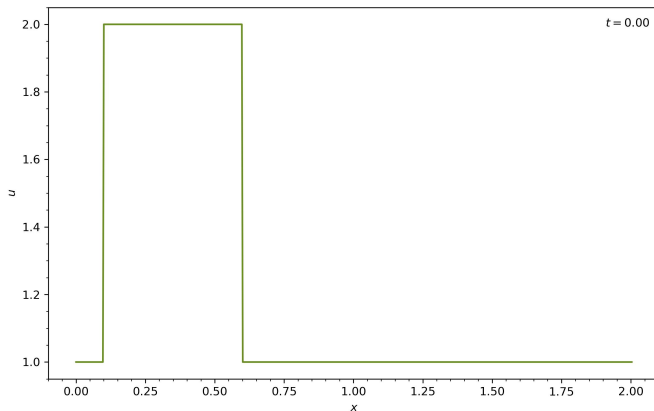
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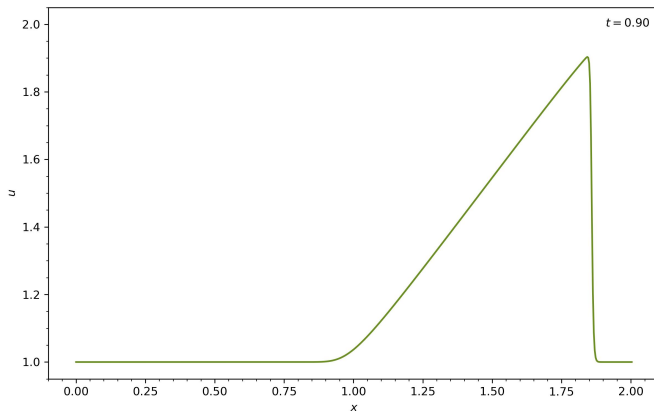
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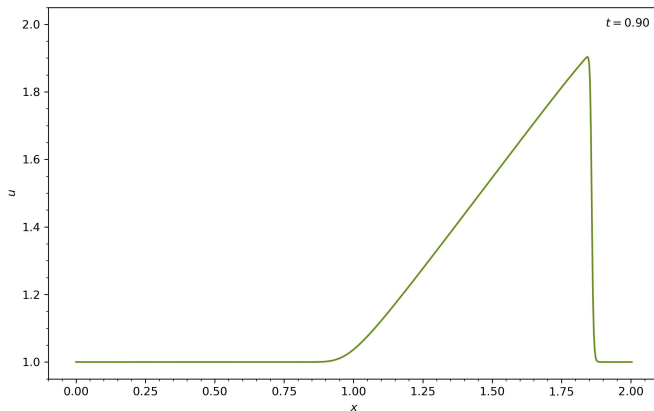
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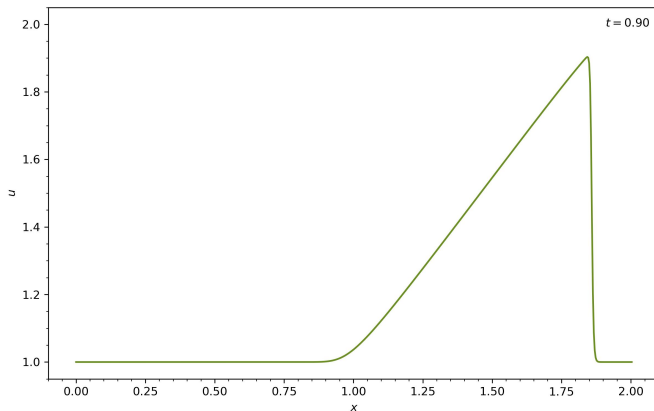
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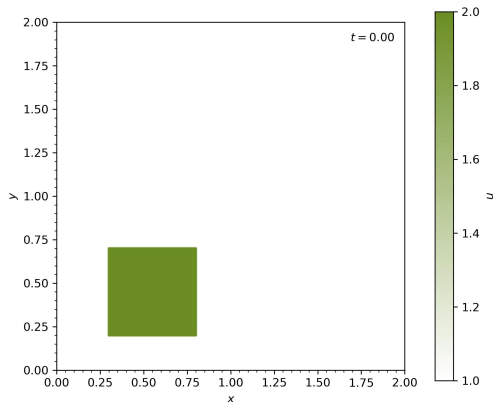
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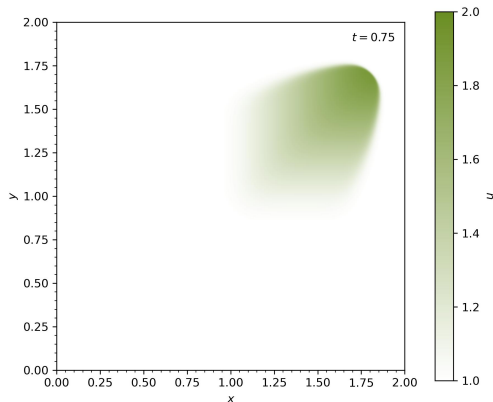
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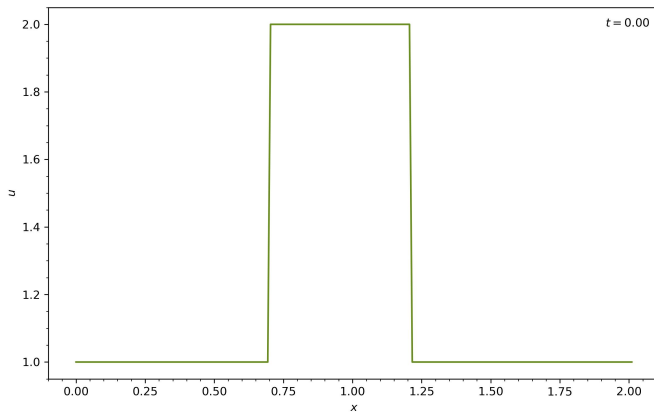
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Diffusion

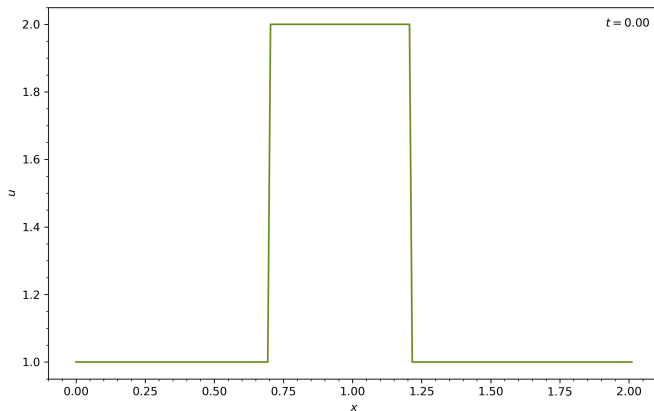
Diffusion

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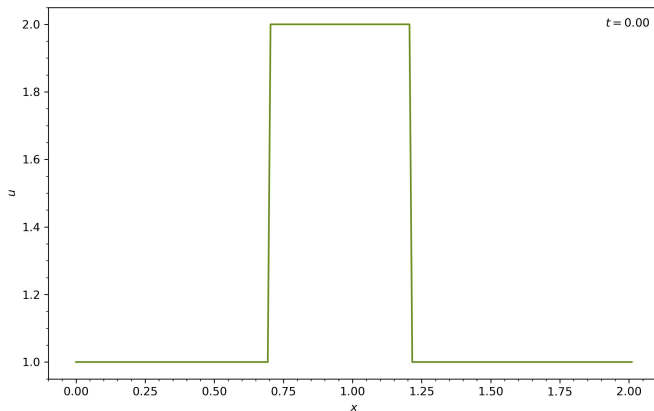
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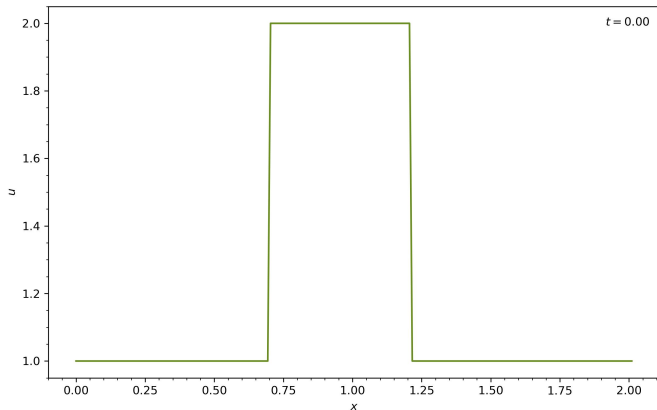
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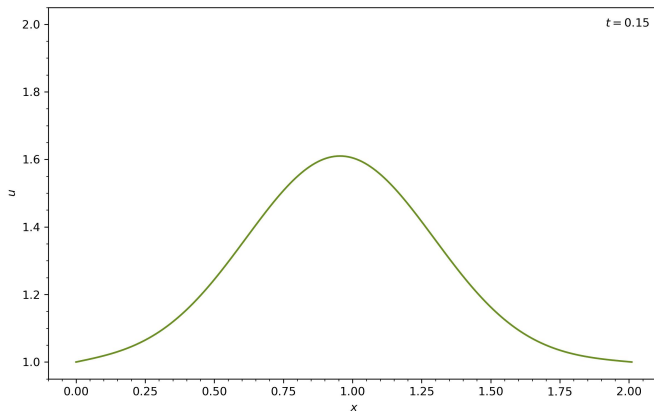
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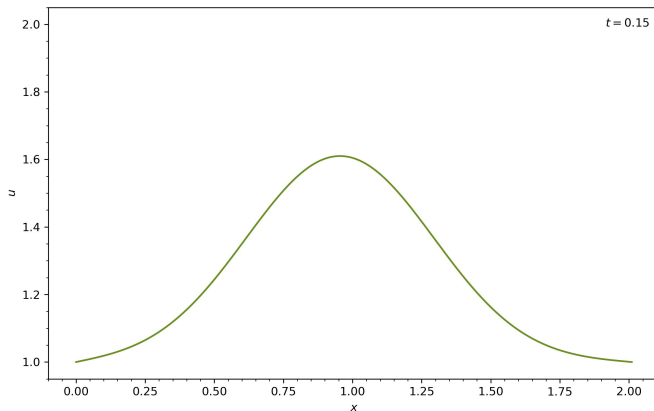
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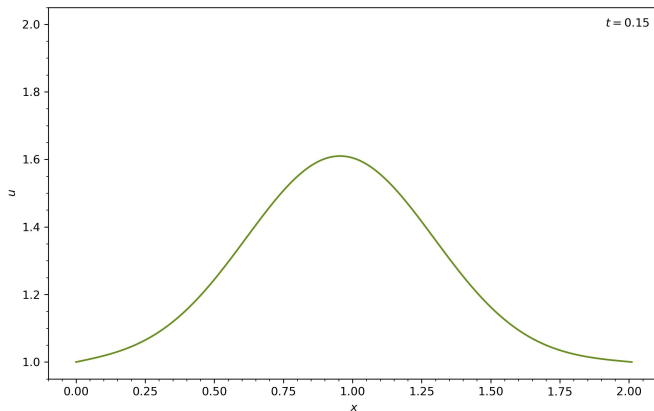
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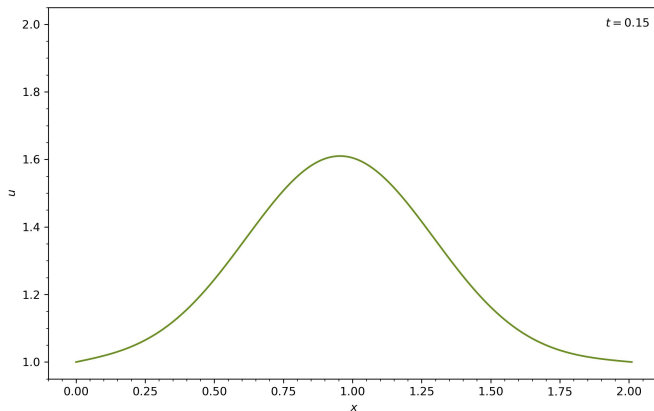
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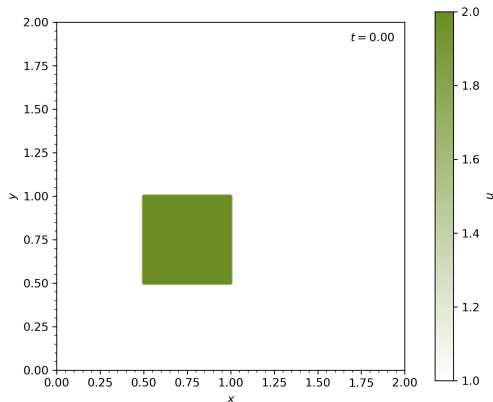
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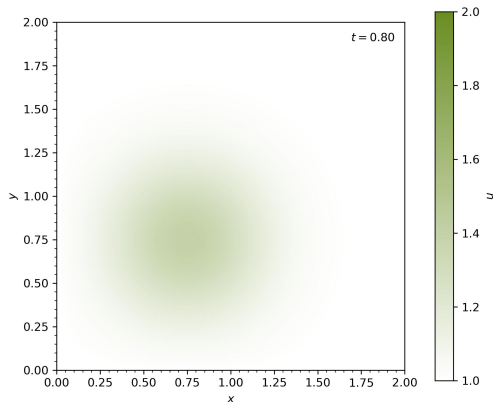
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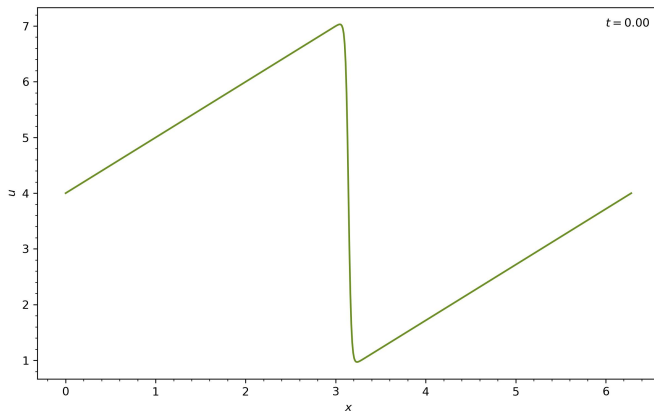
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Bateman-Burgers Equation

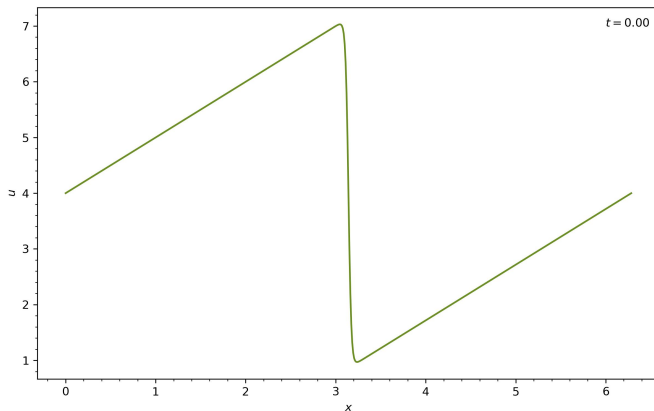
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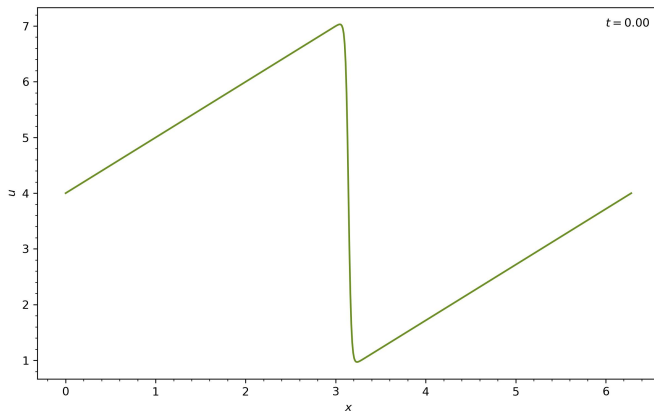
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- Initialization:
 - sawtooth wave



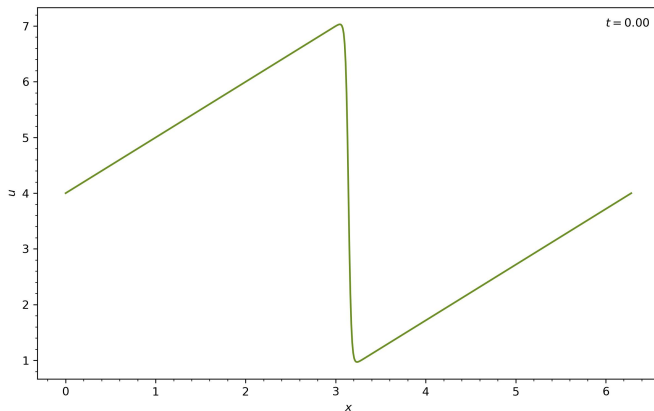
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*



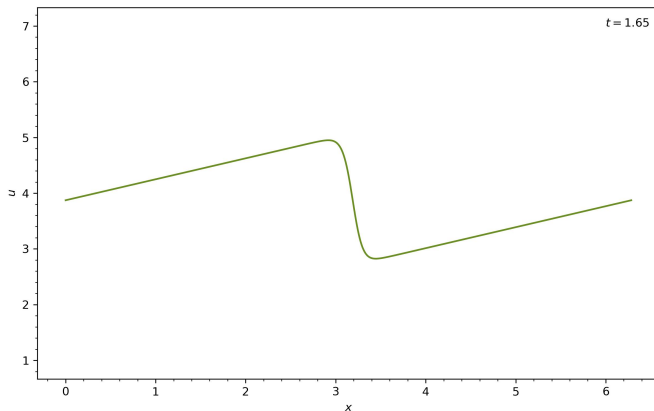
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic



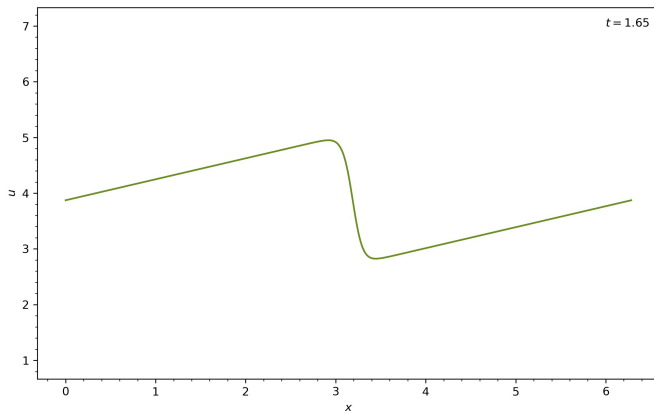
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*



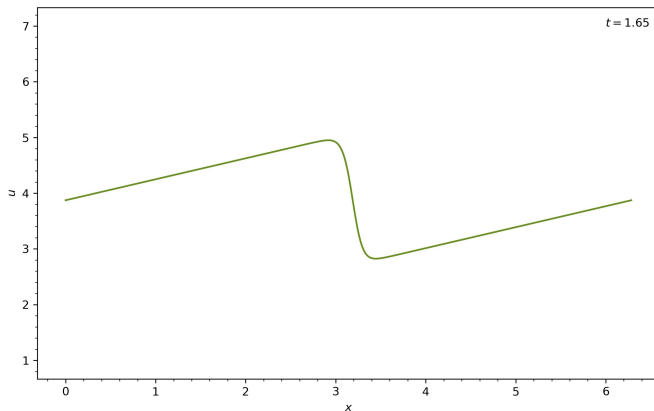
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*
 - nonlinear steepening



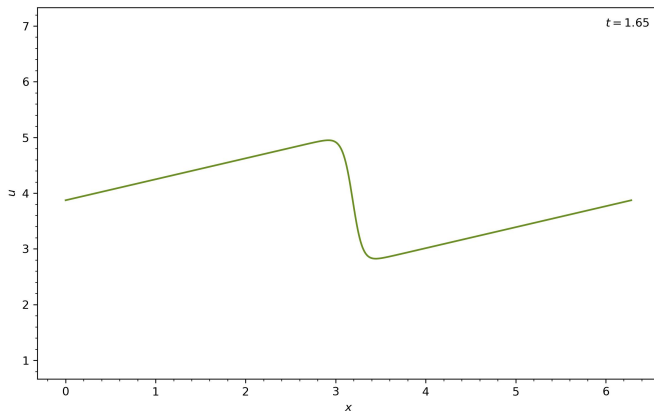
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*
 - nonlinear steepening
 - viscous smoothing



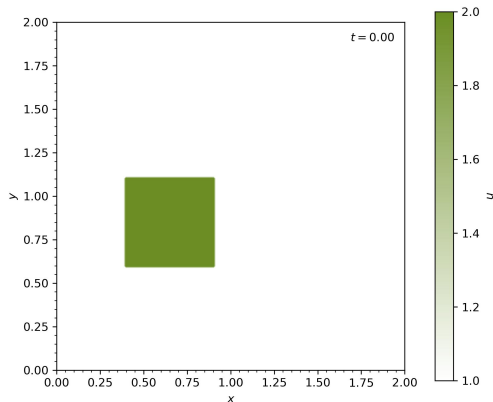
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



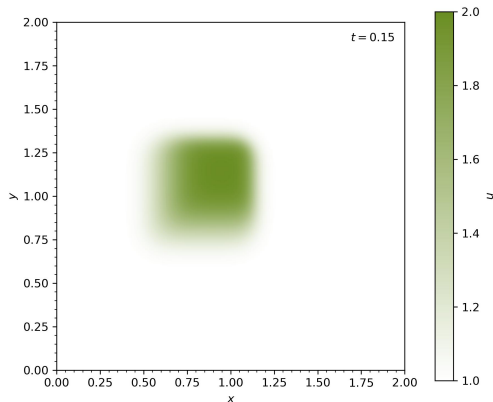
Bateman-Burgers Equation

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



Bateman-Burgers Equation

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



Equilibrium Search

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

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$$\nabla^2 p = 0$$

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- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

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Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- *Gravity*: $\nabla^2 \Phi = 4\pi G \rho$
- *Electrostatics*: $\nabla^2 V = -\rho / \epsilon$

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

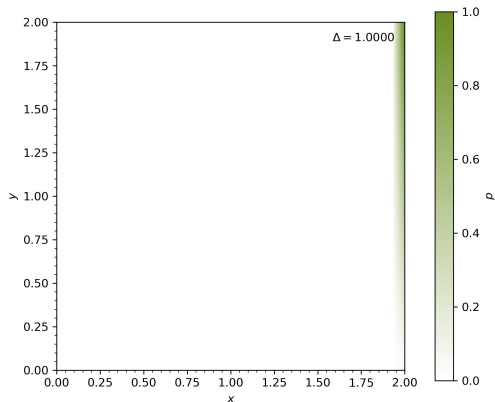
$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G \rho$
- Electrostatics: $\nabla^2 V = -\rho / \epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla) u)$

Laplace Equation

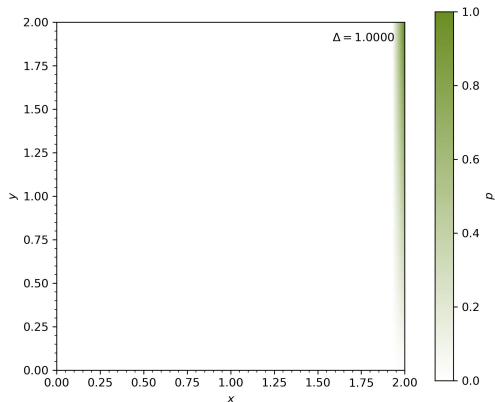
Laplace Equation

■ Initialization:



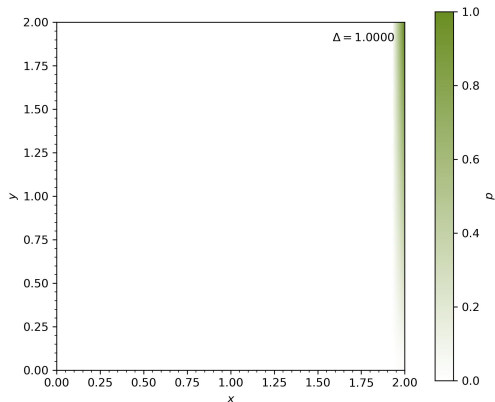
Laplace Equation

- *Initialization:*
 - empty field



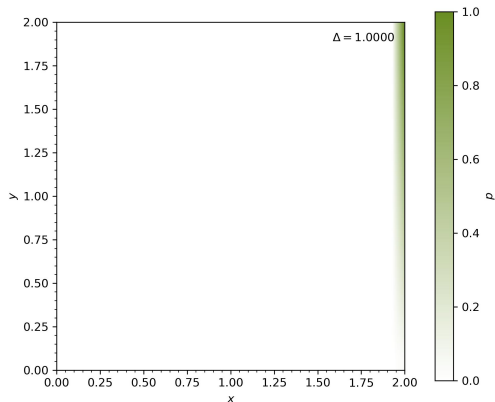
Laplace Equation

- *Initialization:*
 - empty field
- *Boundaries:*



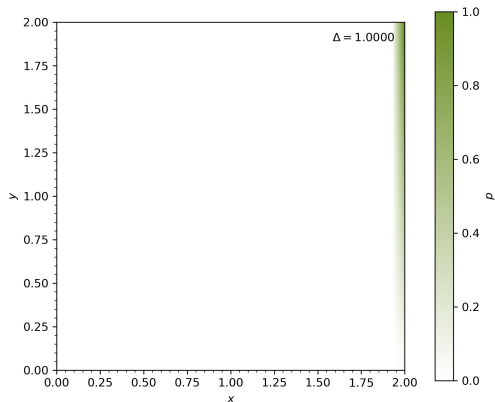
Laplace Equation

- *Initialization:*
 - empty field
- *Boundaries:*
 - *Left:* fixed zero



Laplace Equation

- *Initialization:*
 - empty field
- *Boundaries:*
 - *Left:* fixed zero
 - *Right:* linear slope



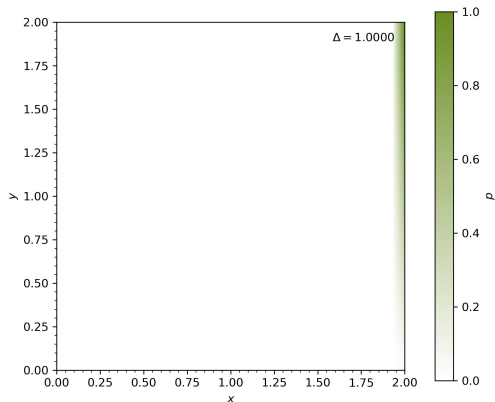
Laplace Equation

- *Initialization:*

- empty field

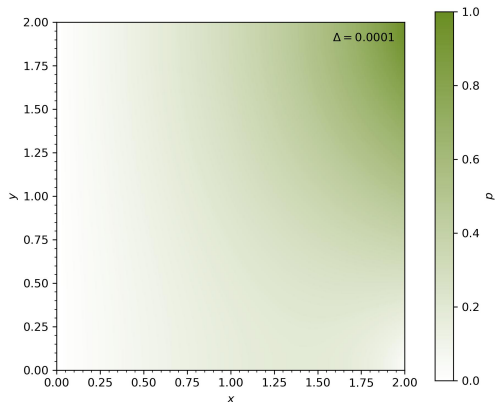
- *Boundaries:*

- *Left:* fixed zero
 - *Right:* linear slope
 - *Top & Bottom:* no gradient



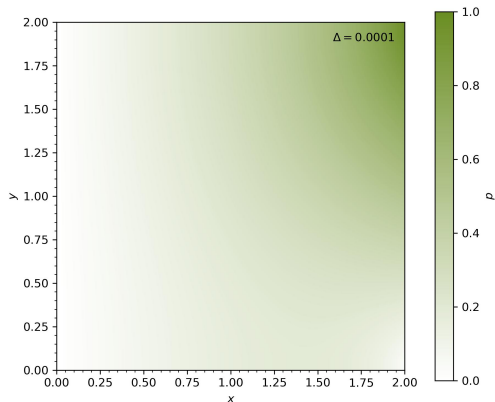
Laplace Equation

- *Initialization:*
 - empty field
- *Boundaries:*
 - *Left:* fixed zero
 - *Right:* linear slope
 - *Top & Bottom:* no gradient
- *Observation:*



Laplace Equation

- *Initialization:*
 - empty field
- *Boundaries:*
 - *Left:* fixed zero
 - *Right:* linear slope
 - *Top & Bottom:* no gradient
- *Observation:*
 - iterative diffusion



Laplace Equation

- *Initialization:*

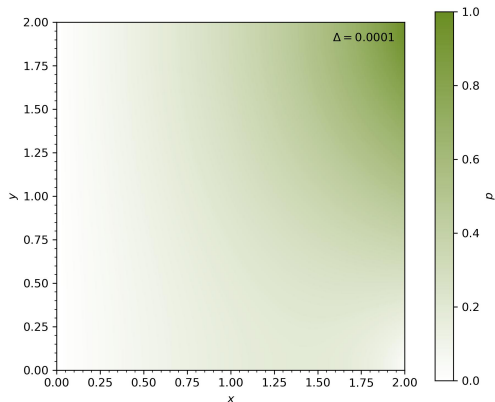
- empty field

- *Boundaries:*

- *Left:* fixed zero
- *Right:* linear slope
- *Top & Bottom:* no gradient

- *Observation:*

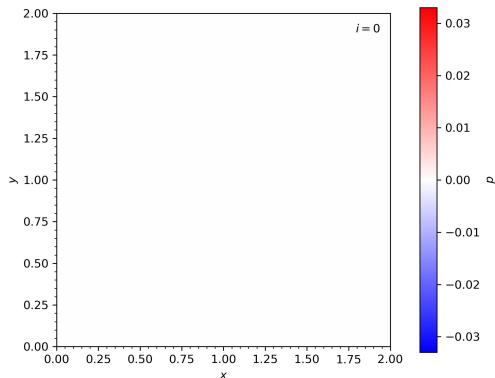
- iterative diffusion
- smooth relaxation



Poisson Equation

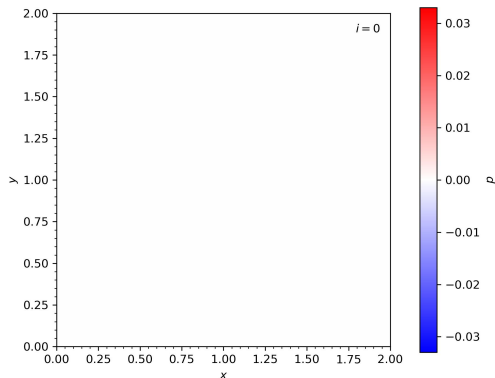
Poisson Equation

■ Initialization:



Poisson Equation

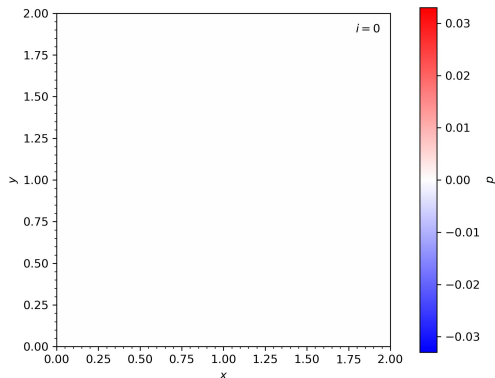
- *Initialization:*
 - empty field



Poisson Equation

■ Initialization:

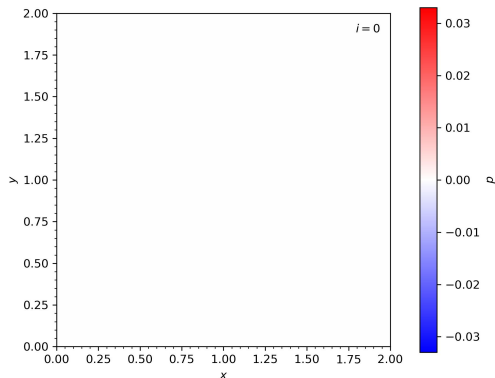
- empty field
- positive source



Poisson Equation

■ Initialization:

- empty field
- positive source
- negative sink

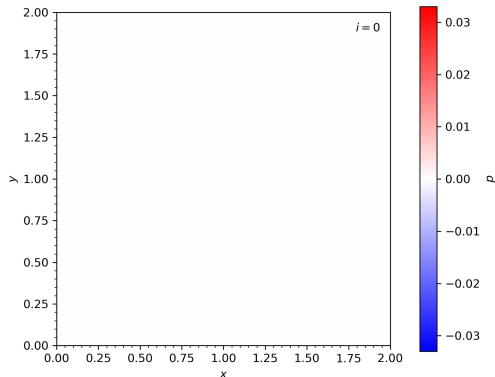


Poisson Equation

- *Initialization:*

- empty field
- positive source
- negative sink

- *Boundaries:*



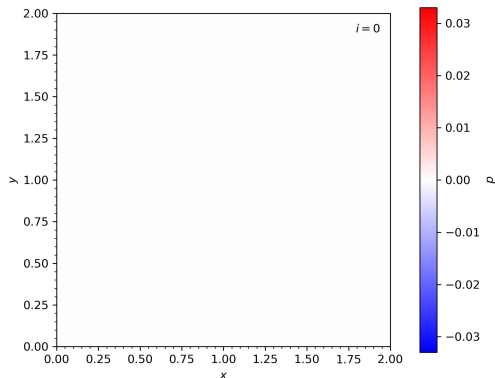
Poisson Equation

- *Initialization:*

- empty field
- positive source
- negative sink

- *Boundaries:*

- All: fixed zero



Poisson Equation

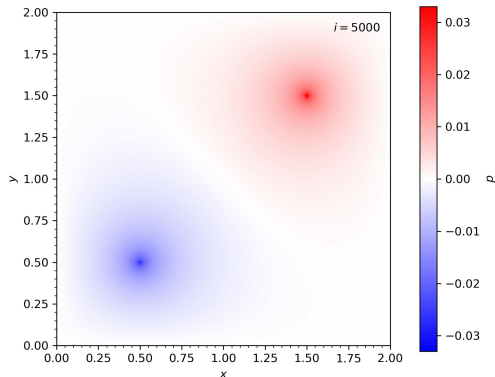
- *Initialization:*

- empty field
- positive source
- negative sink

- *Boundaries:*

- All: fixed zero

- *Observation:*



Poisson Equation

- *Initialization:*

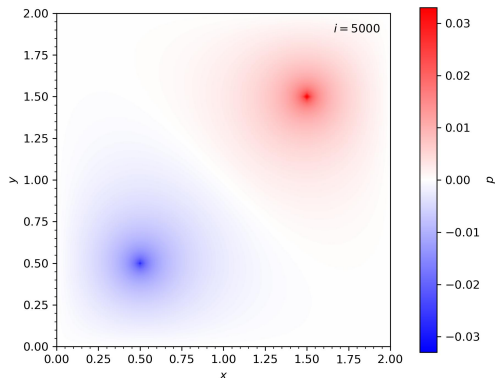
- empty field
- positive source
- negative sink

- *Boundaries:*

- All: fixed zero

- *Observation:*

- dipole field potential



Poisson Equation

- *Initialization:*

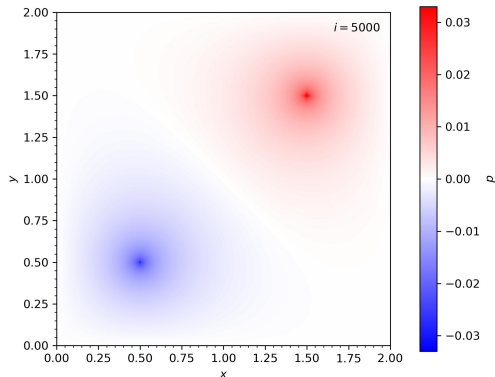
- empty field
- positive source
- negative sink

- *Boundaries:*

- All: fixed zero

- *Observation:*

- dipole field potential
- converging relaxation



Navier-Stokes Equations

Navier-Stokes Equations

- *Momentum Equation*: ensure $\nabla \cdot \mathbf{u} = 0$ by including ∇p to instantaneously cancel out any divergence

Navier-Stokes Equations

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$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

Navier-Stokes Equations

- *Momentum Equation*: ensure $\nabla \cdot \mathbf{u} = 0$ by including ∇p to instantaneously cancel out any divergence

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- *Pressure Poisson*: require p to have relaxed to its equilibrium for all t before projecting

Navier-Stokes Equations

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$$\nabla^2 p = \rho \nabla \cdot ((\mathbf{u} \cdot \nabla) \mathbf{u})$$

Navier-Stokes Equations

- *Momentum Equation*: ensure $\nabla \cdot \mathbf{u} = 0$ by including ∇p to instantaneously cancel out any divergence

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

- *Pressure Poisson*: require p to have relaxed to its equilibrium for all t before projecting

$$\nabla^2 p = \rho \nabla \cdot ((\mathbf{u} \cdot \nabla) \mathbf{u})$$

- fulfill at each step to accurately evolve initial configuration under given boundary conditions

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

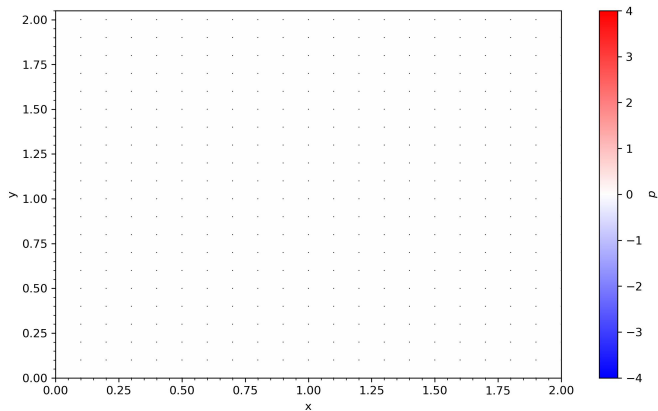
Rayleigh-Taylor Instability

Outlook

Lid Driven Cavity Flow

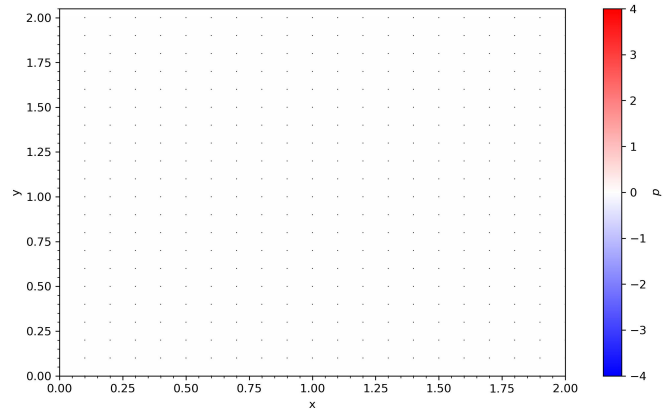
Lid Driven Cavity Flow

■ Initialization:



Lid Driven Cavity Flow

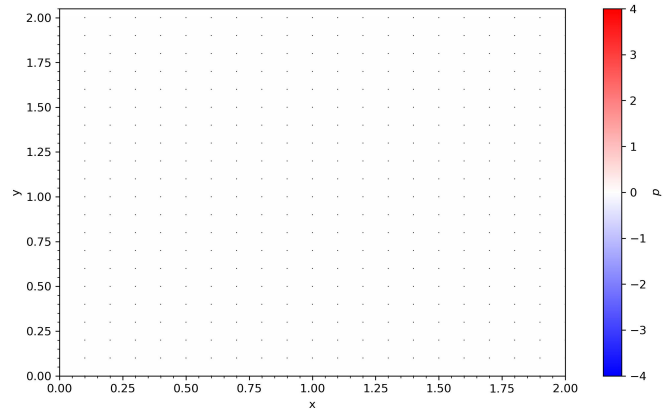
- *Initialization:*
 - vanishing velocity



Lid Driven Cavity Flow

■ Initialization:

- vanishing velocity
- zero pressure

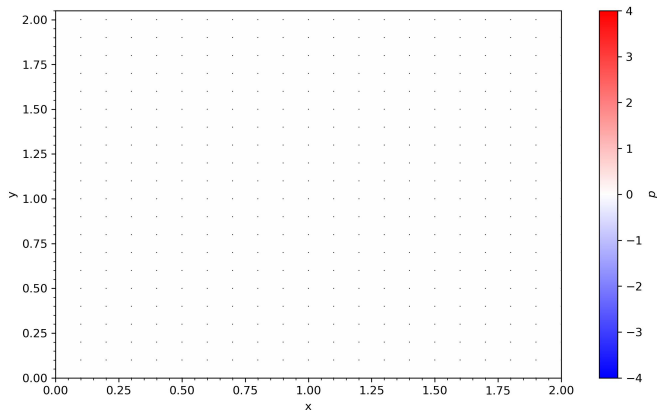


Lid Driven Cavity Flow

- *Initialization:*

- vanishing velocity
- zero pressure

- *Boundaries:*



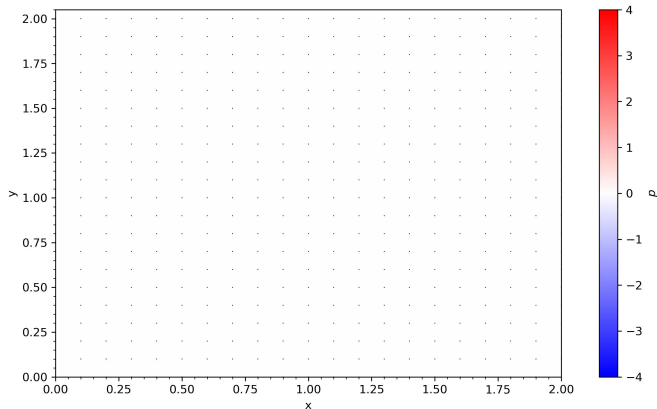
Lid Driven Cavity Flow

- *Initialization:*

- vanishing velocity
- zero pressure

- *Boundaries:*

- *Top:* $u = 1$ & $\nabla p = 0$



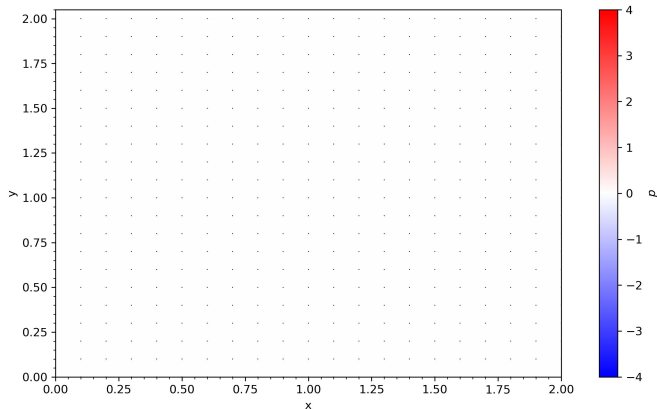
Lid Driven Cavity Flow

- *Initialization:*

- vanishing velocity
- zero pressure

- *Boundaries:*

- *Top:* $u = 1$ & $\nabla p = 0$
- *Bottom:* $u = 0$ & $p = 0$



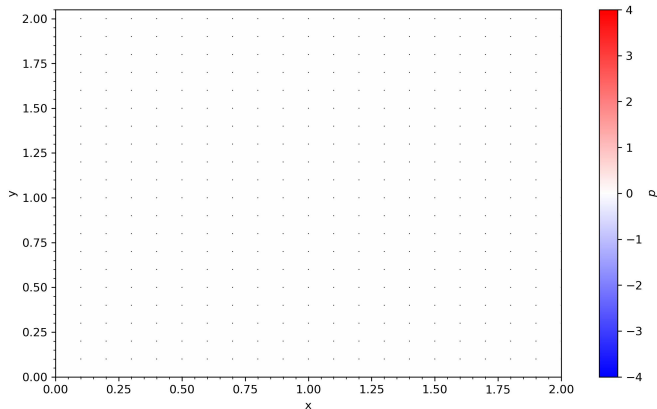
Lid Driven Cavity Flow

■ Initialization:

- vanishing velocity
- zero pressure

■ Boundaries:

- Top: $u = 1$ & $\nabla p = 0$
- Bottom: $u = 0$ & $p = 0$
- Left & Right: $u = 0$ & $\nabla p = 0$



Lid Driven Cavity Flow

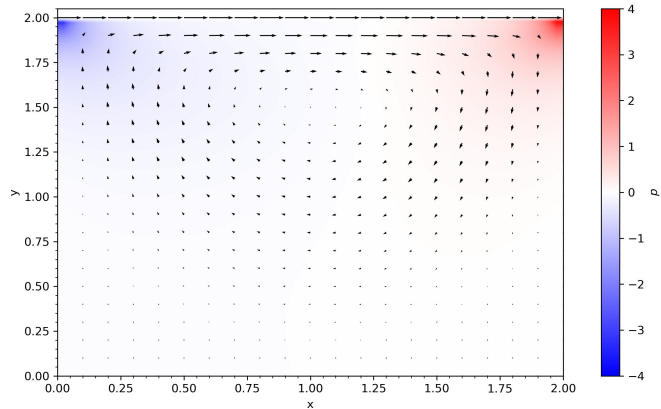
- *Initialization:*

- vanishing velocity
- zero pressure

- *Boundaries:*

- *Top:* $u = 1$ & $\nabla p = 0$
- *Bottom:* $u = 0$ & $p = 0$
- *Left & Right:* $u = 0$ & $\nabla p = 0$

- *Observation:*



Lid Driven Cavity Flow

■ Initialization:

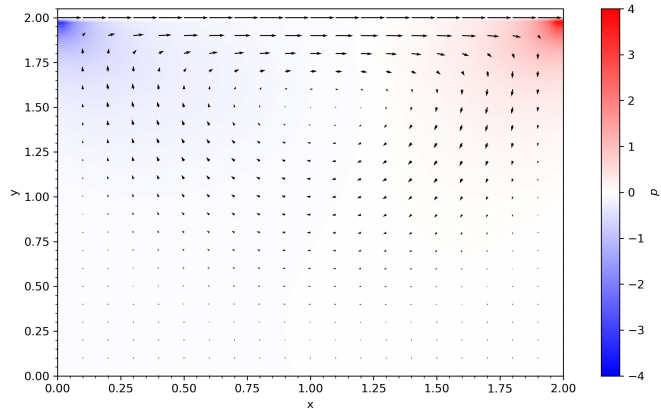
- vanishing velocity
- zero pressure

■ Boundaries:

- Top: $u = 1$ & $\nabla p = 0$
- Bottom: $u = 0$ & $p = 0$
- Left & Right: $u = 0$ & $\nabla p = 0$

■ Observation:

- vortex formation



Lid Driven Cavity Flow

■ Initialization:

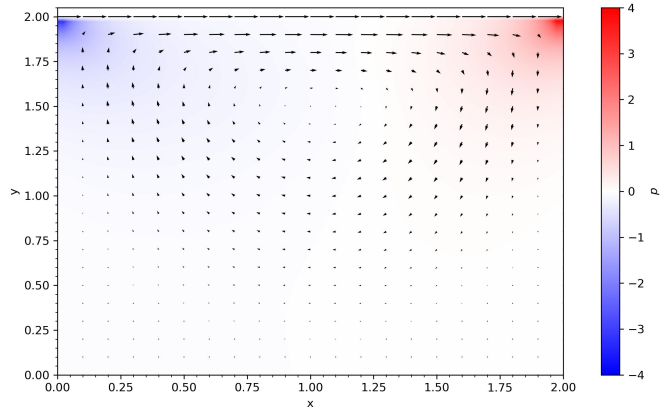
- vanishing velocity
- zero pressure

■ Boundaries:

- Top: $u = 1$ & $\nabla p = 0$
- Bottom: $u = 0$ & $p = 0$
- Left & Right: $u = 0$ & $\nabla p = 0$

■ Observation:

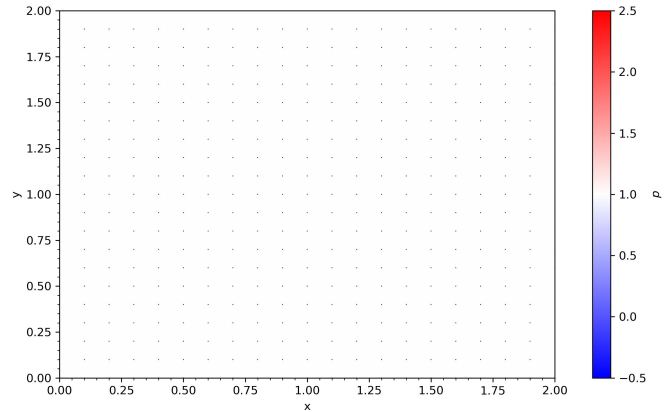
- vortex formation
- pressure gradient



Pressure Driven Channel Flow

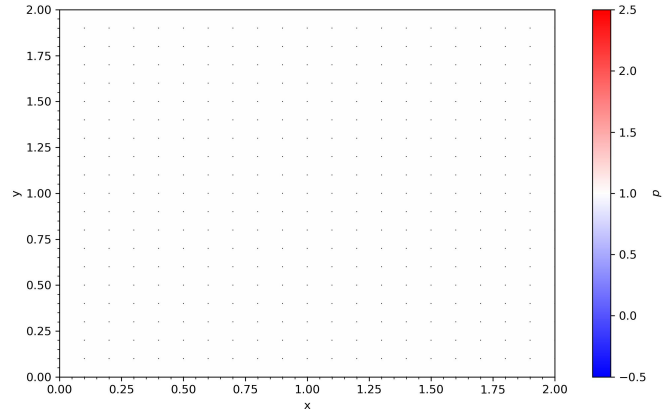
Pressure Driven Channel Flow

■ Initialization:



Pressure Driven Channel Flow

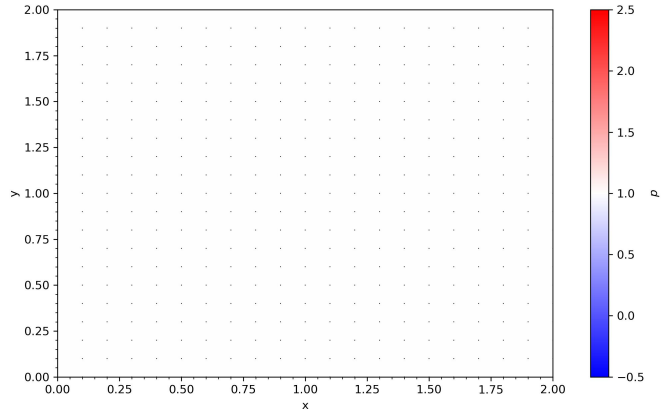
- *Initialization:*
 - vanishing velocity



Pressure Driven Channel Flow

■ Initialization:

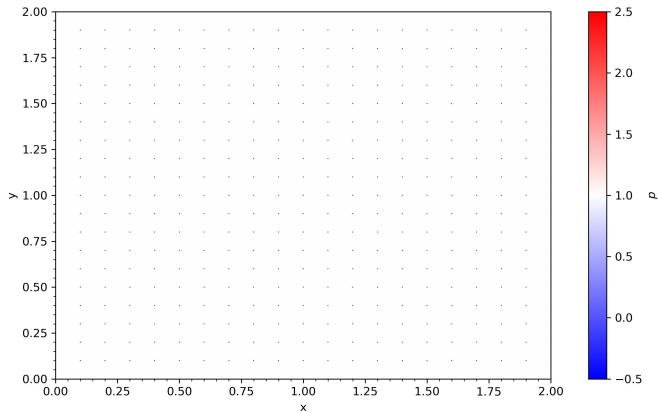
- vanishing velocity
- unity pressure



Pressure Driven Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- *Poiseuille*: constant source

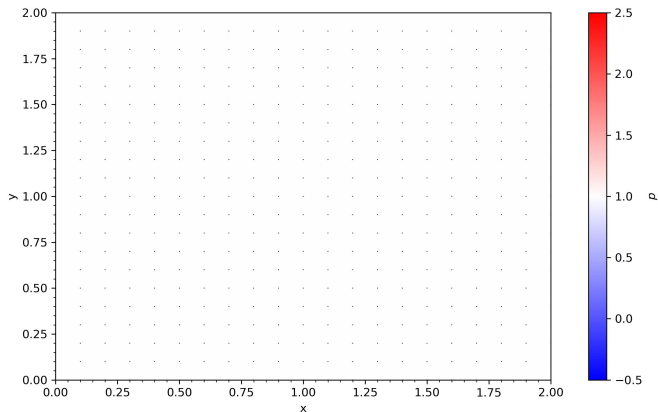


Pressure Driven Channel Flow

- *Initialization:*

- vanishing velocity
- unity pressure
- *Poiseuille*: constant source

- *Boundaries:*



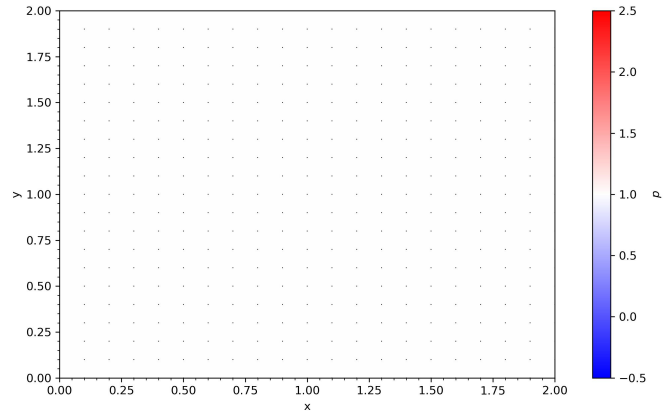
Pressure Driven Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- *Poiseuille*: constant source

■ Boundaries:

- *Top & Bottom*: $u = 0$ & $\nabla p = 0$



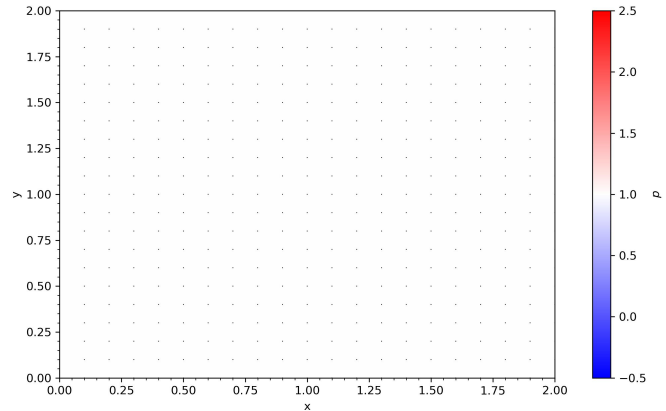
Pressure Driven Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- *Poiseuille*: constant source

■ Boundaries:

- *Top & Bottom*: $u = 0$ & $\nabla p = 0$
- *Left & Right*: periodic



Pressure Driven Channel Flow

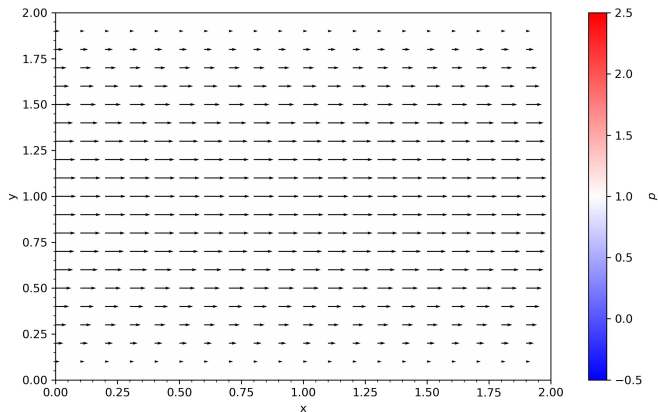
■ Initialization:

- vanishing velocity
- unity pressure
- Poiseuille: constant source

■ Boundaries:

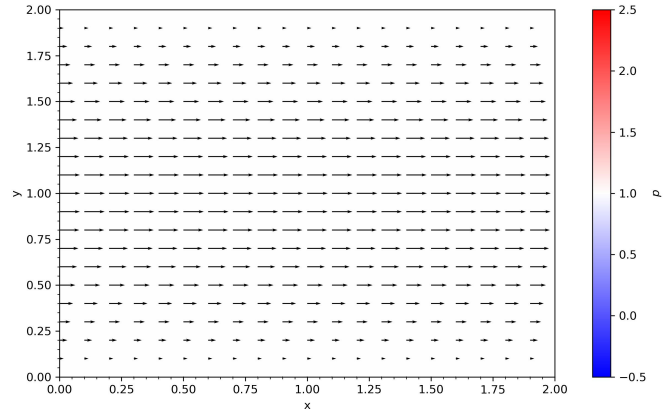
- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:



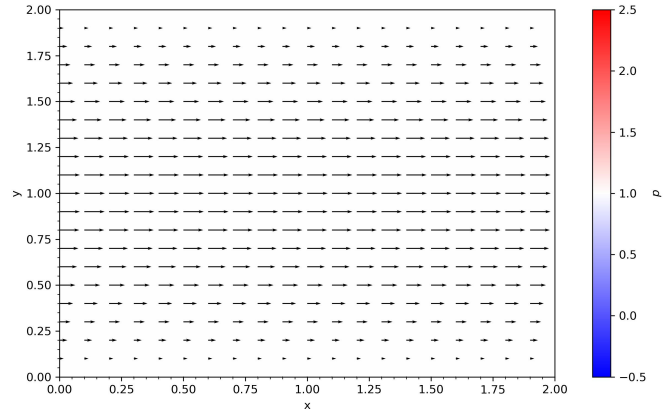
Pressure Driven Channel Flow

- *Initialization:*
 - vanishing velocity
 - unity pressure
 - *Poiseuille:* constant source
- *Boundaries:*
 - *Top & Bottom:* $u = 0$ & $\nabla p = 0$
 - *Left & Right:* periodic
- *Observation:*
 - curved velocity profile



Pressure Driven Channel Flow

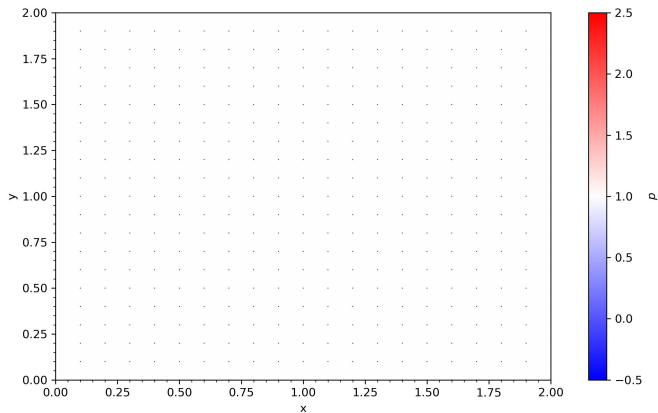
- *Initialization:*
 - vanishing velocity
 - unity pressure
 - *Poiseuille*: constant source
- *Boundaries:*
 - *Top & Bottom*: $u = 0$ & $\nabla p = 0$
 - *Left & Right*: periodic
- *Observation:*
 - curved velocity profile
 - boundary layer formation



Locally Forced Channel Flow

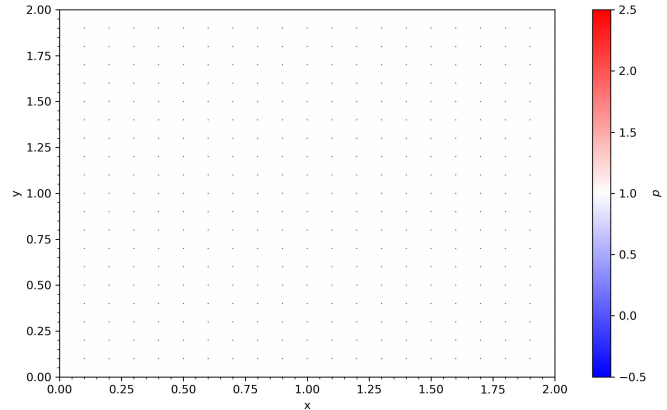
Locally Forced Channel Flow

■ Initialization:



Locally Forced Channel Flow

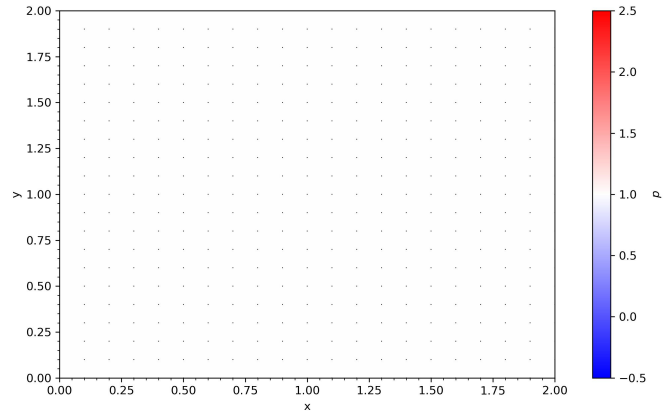
- *Initialization:*
 - vanishing velocity



Locally Forced Channel Flow

■ Initialization:

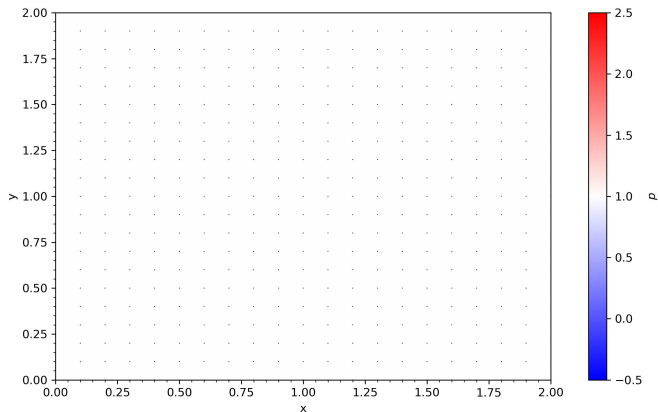
- vanishing velocity
- unity pressure



Locally Forced Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- rectangular source

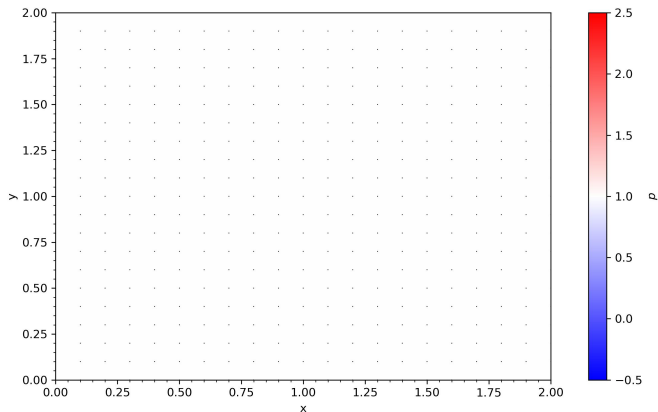


Locally Forced Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:



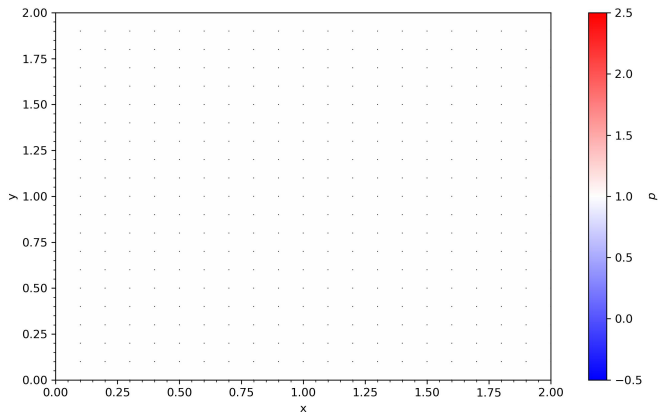
Locally Forced Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$



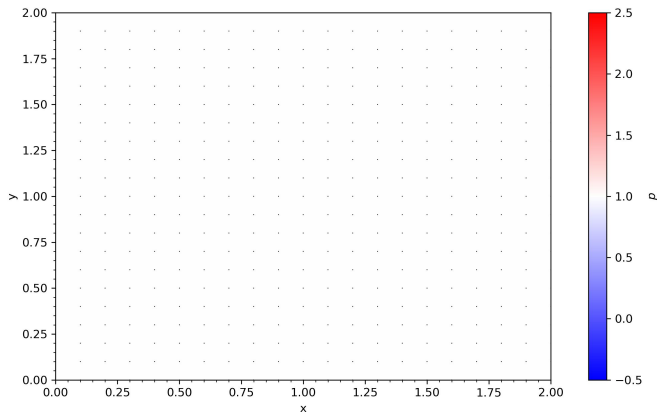
Locally Forced Channel Flow

■ Initialization:

- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic



Locally Forced Channel Flow

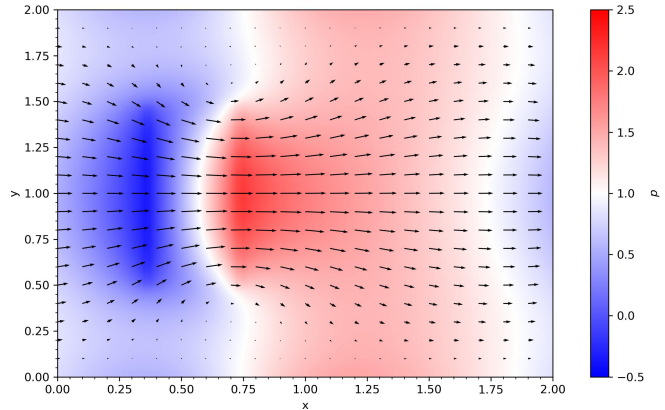
■ Initialization:

- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:



Locally Forced Channel Flow

■ Initialization:

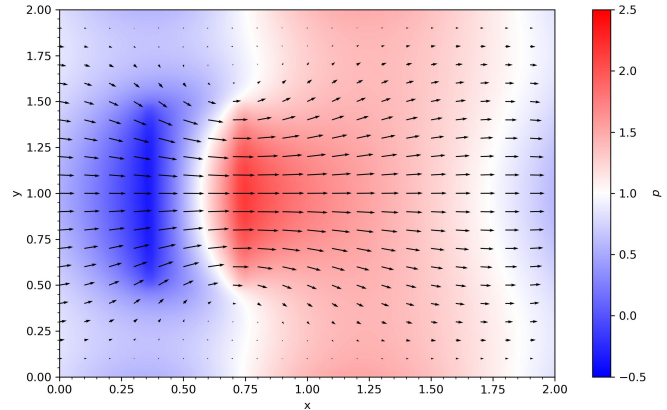
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■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:

- central jet formation



Locally Forced Channel Flow

■ Initialization:

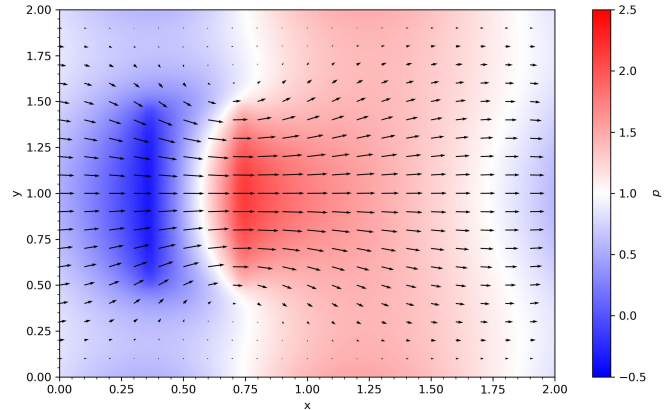
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 - mass and energy can leak due to numerical errors

Integration Techniques

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- *Spatial:*

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Discretized Formulation

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■ *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$

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- Idea: calculate flux through faces of control volumes that divide the domain using integral form

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 - oppositely directed horizontal velocity layers

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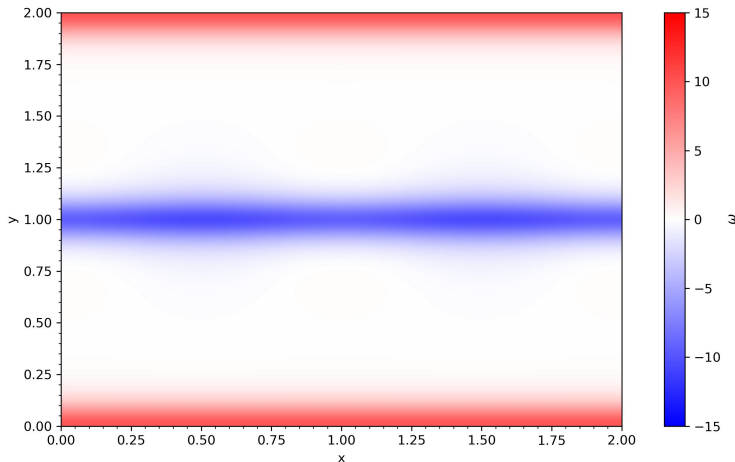
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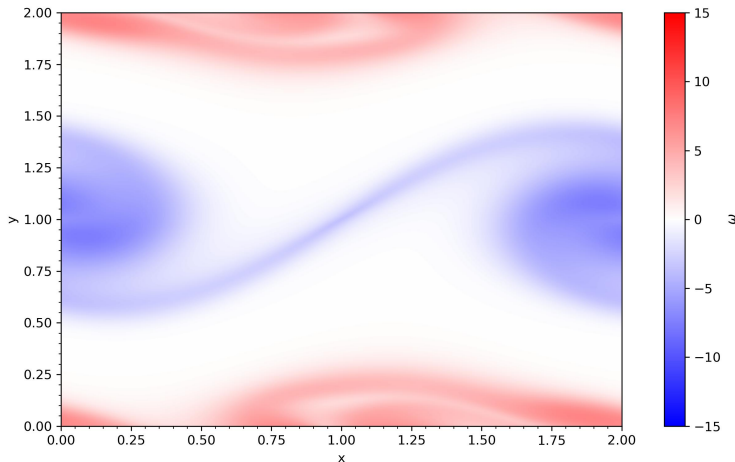
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- mixing occurs and creates turbulence

Visualization

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Visualization



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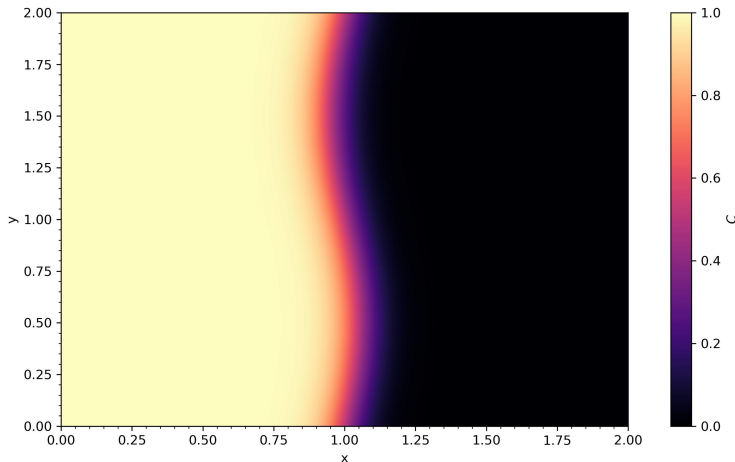
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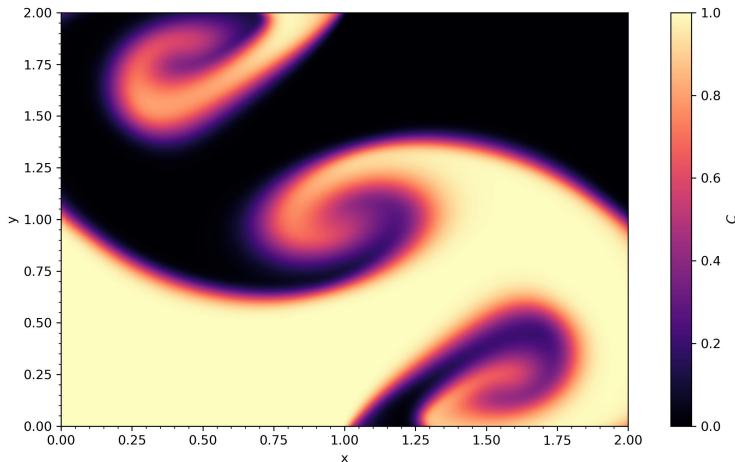
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- Kelvin-Helmholtz Instabilities: spiraling vortices from friction lead to turbulent mixing

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■ Applications:

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| ■ <i>Plasma</i> : use magnetohydrodynamics | ■ <i>Degeneracy</i> : need quantum effects |
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Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)